

Dev Patel

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EDUCATION

University of Central Florida | Orlando, Florida

Aug. 2026

B.S. in Computer Science - Awards and Activities: Dean's List, AI@UCF, KnightHacks

EXPERIENCE

United Financial Services - Vitruvix | Sarasota, Florida

Aug. 2026 - Present

Software Engineer / AI Engineer

- Engineered a production LLM validation pipeline using Python, Agno, LiteLLM, and Pydantic, combining structured outputs, deterministic guardrails, and fallback logic to improve reliability of self-hosted LLM responses.
- Analyzed 730+ model-generated outputs to identify reasoning-leak patterns, correcting an overstated ~17% issue metric to ~3% and converting findings into stronger validation and safety rules.
- Improved real-time AI workflow performance by parallelizing up to 4 LLM evaluations with asyncio and 15-second timeouts, reducing blocking from minute-scale worst cases to seconds while preserving streamed application behavior

SwapVest Capital | Tampa, Florida

Apr. 2025 - Jan. 2026

Founding Software Developer / Fintech Software Startup

- Engineered a peer-to-peer equity swap platform (FastAPI, React, PostgreSQL) with 37 REST + SSE endpoints, a 14-table relational schema, row-level locking on concurrent cash mutations, and JWT authentication.
- Engineered a Decimal-based swap valuation engine in FastAPI to calculate dual-leg mark-to-market values, collateral requirements, and zero-sum P&L updates for simulated peer-to-peer stock exposure swaps..
- Implemented SSE-based live pricing (JWT via query param; 5-state reconnect machine; polling fallback) and an asyncio scheduler that auto-settles matured contracts with double-entry cash P&L transfer.

PROJECTS

CARVER MATRICES | (USASOC sponsored Senior Design Project)

Jan. 2026

Role: Full-Stack Developer

- Developed a Tauri/Rust desktop application for USASOC CARVER threat-assessment workflows, supporting air-gapped deployment with an embedded llama.cpp AI sidecar for local scoring recommendations, custom PDF generation and import, and Argon2id-secured user authentication.
- Built 26 typed Tauri IPC commands in Rust across 6 domains, enforcing per-command session guards and structured error propagation across the full application.
- Designed a normalized SQLite schema with foreign key constraints, score validation, cascading deletes, and idempotent migrations to support persistent storage for users, operations, matrices, and evidence annotations

Sports Prediction Supervised Learning Model

Aug. 2025

Personal Project

- Developing a full-stack sports analytics platform using React, Django REST Framework, PostgreSQL, pandas, and scikit-learn to compare historical NBA teams across 25+ seasons and generate stat-based win-probability predictions.
- Built a Python/pandas ETL pipeline to transform 20,000+ raw NBA box-score records into 750+ team-season profiles across offensive, defensive, shooting, rebounding, pace, and efficiency metrics.
- Engineered 40+ difference-based matchup features and implemented a scikit-learn prediction engine served through Django REST APIs and visualized in a React statistical comparison dashboard.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL, TypeScript, Rust, HTML, CSS

Frameworks/Libraries: React, FastAPI, Django, Tailwind CSS, Express.js, Node.js, scikit-learn

Databases: PostgreSQL, SQLite, MongoDB, MySQL

Developer Tools: Docker, AWS, Git, GitHub, Jira, Agile, Figma, CI/CD, Postman, Unit Testing, Claude Code